
Topic 11

Measurement, reliability, and
descriptive research

Overview

- Measurement scales
- Reliability
- Descriptive research

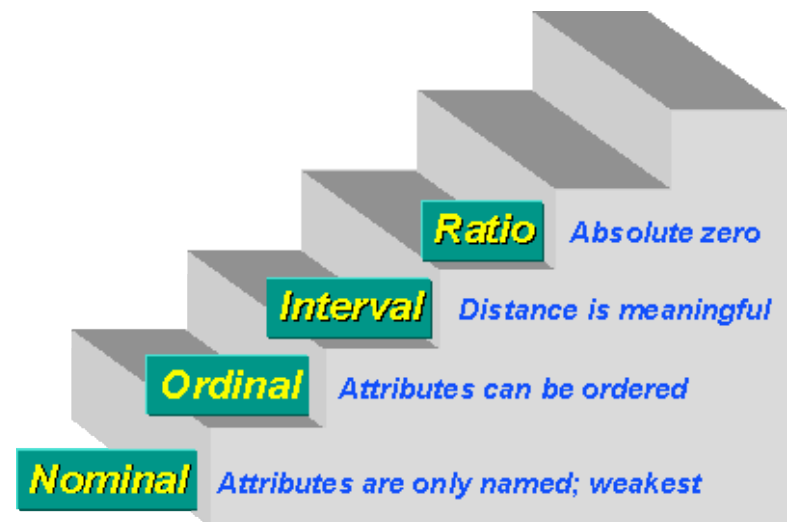
Measurement scales

- Measurement
 - A systematic way to assign numbers or names to objects and their features



Measurement scales

- Measurement scales
 - Nominal scale - differences (categorical)
 - Ordinal scale - ranked order
 - Interval scale - equal intervals (add/subtract)
 - Ratio scale - absolute zero (ratio's)



Measurement scales

- Measurement scales
 - Nominal scale - differences (categorical)
 - Ordinal scale - ranked order
 - Interval scale - equal intervals (add/subtract)
 - Ratio scale - absolute zero (ratio's)
- Importance of measurement scales
 - determines the possible statistical operations
 - determines the possible conclusions

Measurement scales

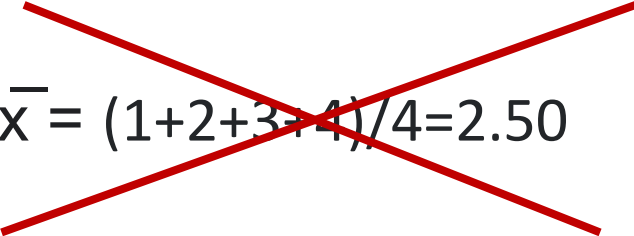
- The board of the VU student restaurant likes to know which meals are preferred by students. To investigate this, a study is performed in which students indicated the extent to which they enjoy specific meals (questionnaire using 5-point scales, with 1 corresponding to very distasteful and 5 corresponding to very delicious). The results showed that spinach pie had an average score of 4 and spaghetti had an average score of 2. It was concluded that students like spinach pie twice as much as spaghetti. Is this conclusion correct?

R1 - 1

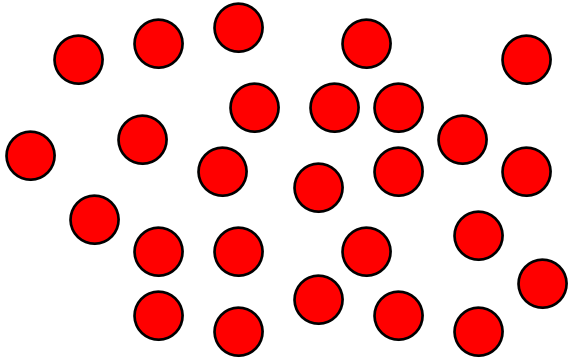
R2 - 2

R3 - 3

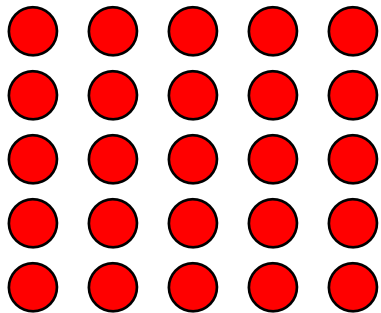
R4 - 4


$$\bar{x} = (1+2+3+4)/4 = 2.50$$

Reliability



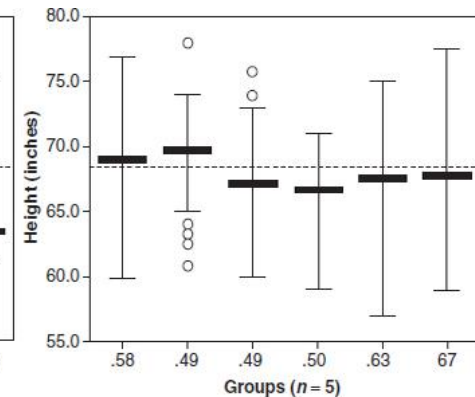
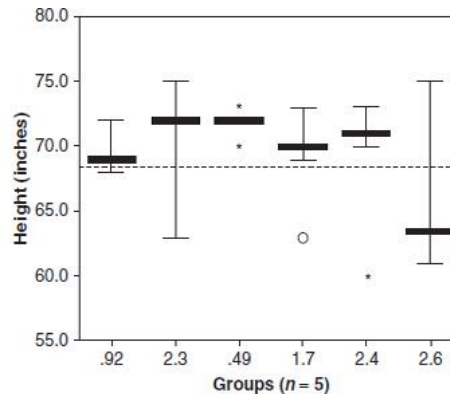
How many red circles do you count?



How many red circles do you count?

Reliability

- Observed score = true score + error score
 - Observed score
 - the measured value
 - True score
 - the real value
 - Error score
 - the difference between the measured value and the real value which is affected by
 - the test (situation)
 - the person



Note: The population mean = 67.7 inches tall (dashed line).

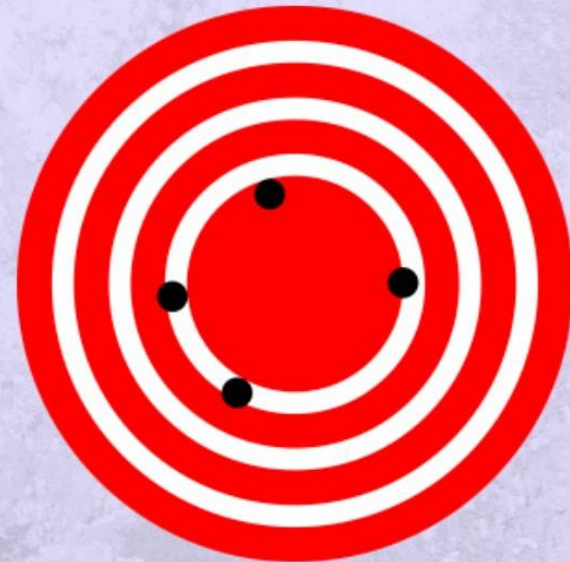
Reliability

Systematic Error



- Using a stretched measuring tape
- Scale that always reads too high or low
- Reading an indicator from a poor angle

Random Error



- Timing depends on reaction time
- People take turns taking readings
- Rounding values up or down.

Reliability

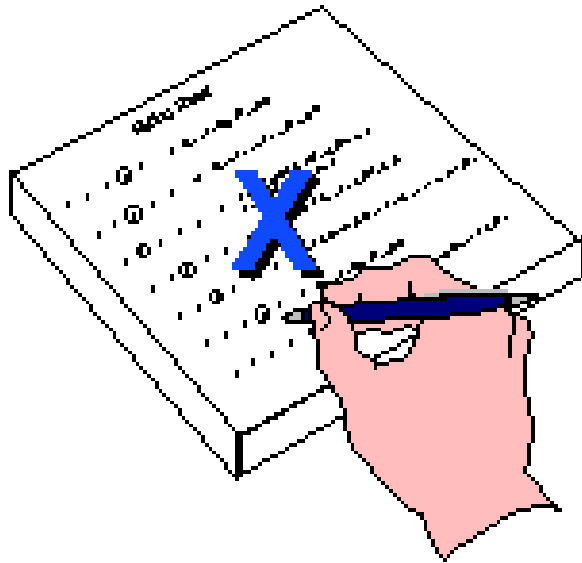
Observed
Score

=

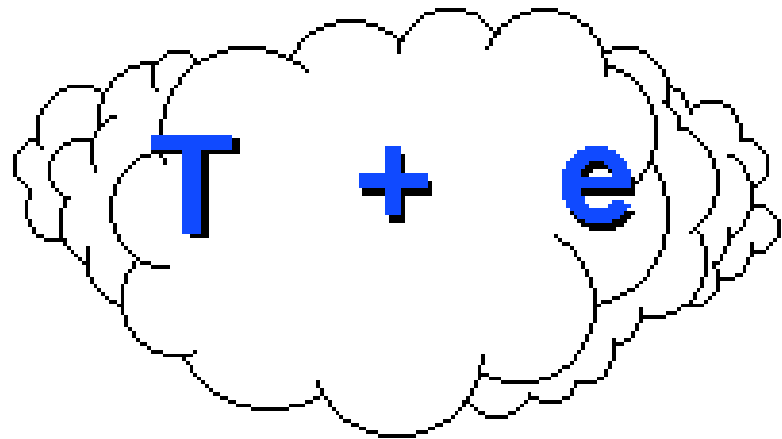
True
Ability

+

Random
Error

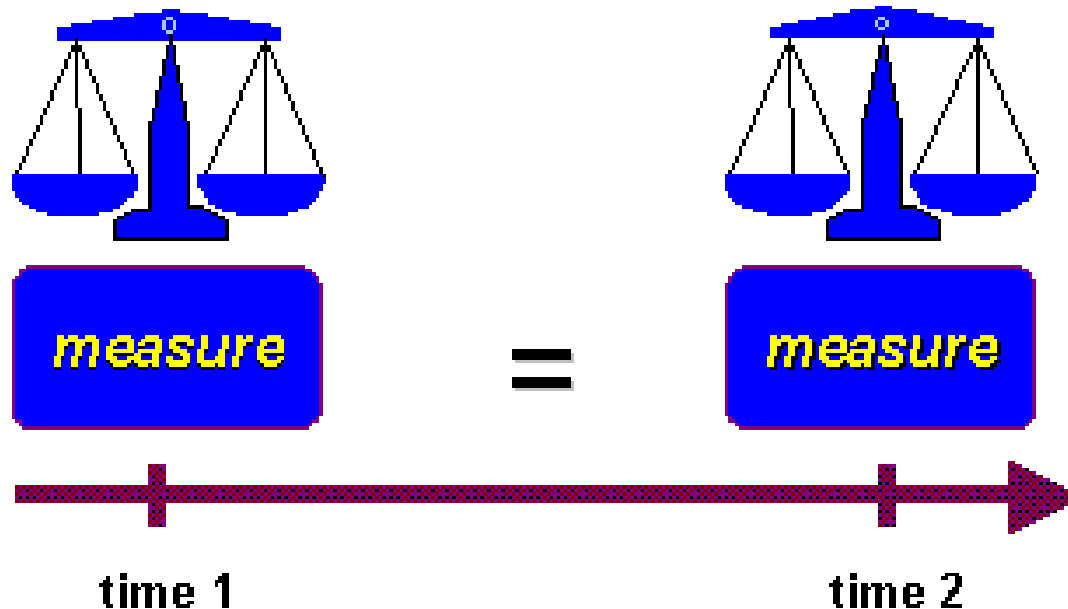


=

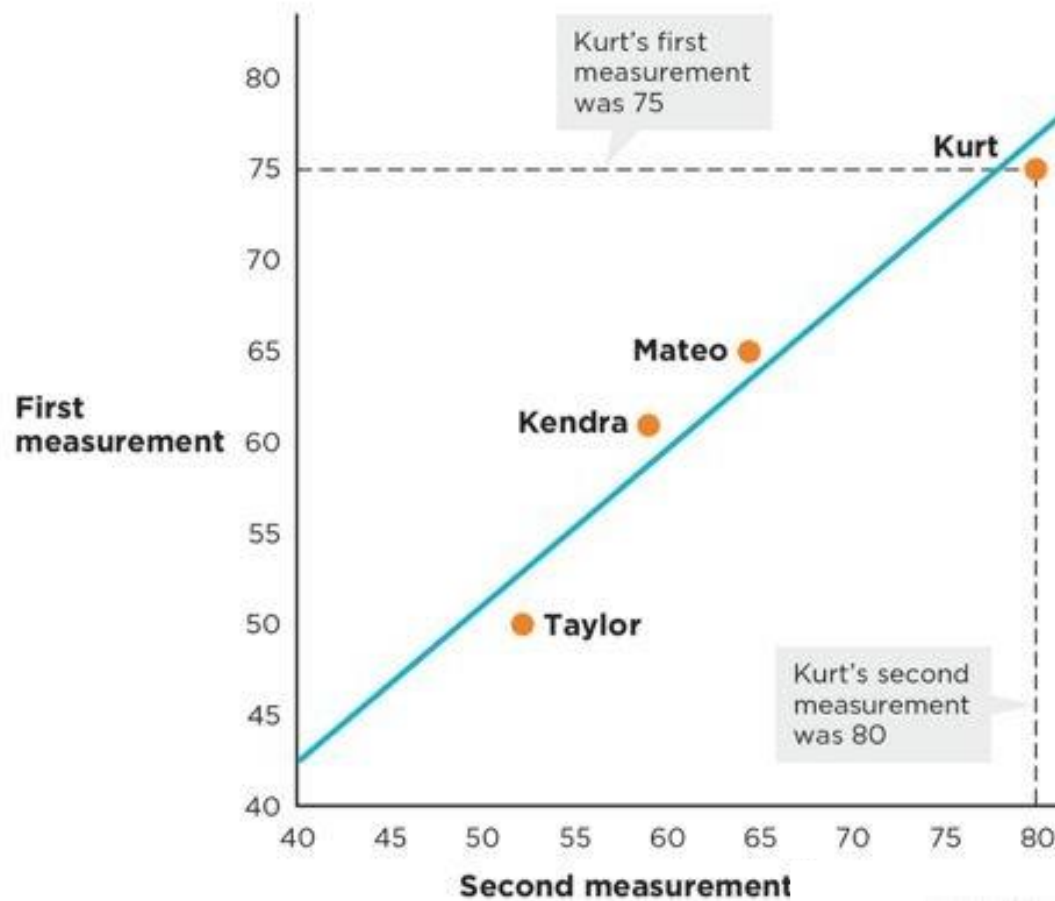


Reliability

- Reliability
 - Consistency of measurements
- Different types of reliability
 - Test reliability
 - Test-retest reliability



Reliability



Reliability

- Demo: retrospective-self-report test
 - *What is the amount of time you spent studying the previous week's chapter?*

In graduate school, my professors emphasized the inaccuracy of retrospective self-report data. However, the truth did not come alive for me until I was doing research on student learning for my PhD thesis. In one study, I asked students to estimate (to the nearest half hour) *the amount of time they spent studying the previous week's chapter*. All my students agreed to answer this question. Nobody said "I don't remember" or "I cannot make that estimate." I assumed the information was accurate, and I prepared to use it in my research.

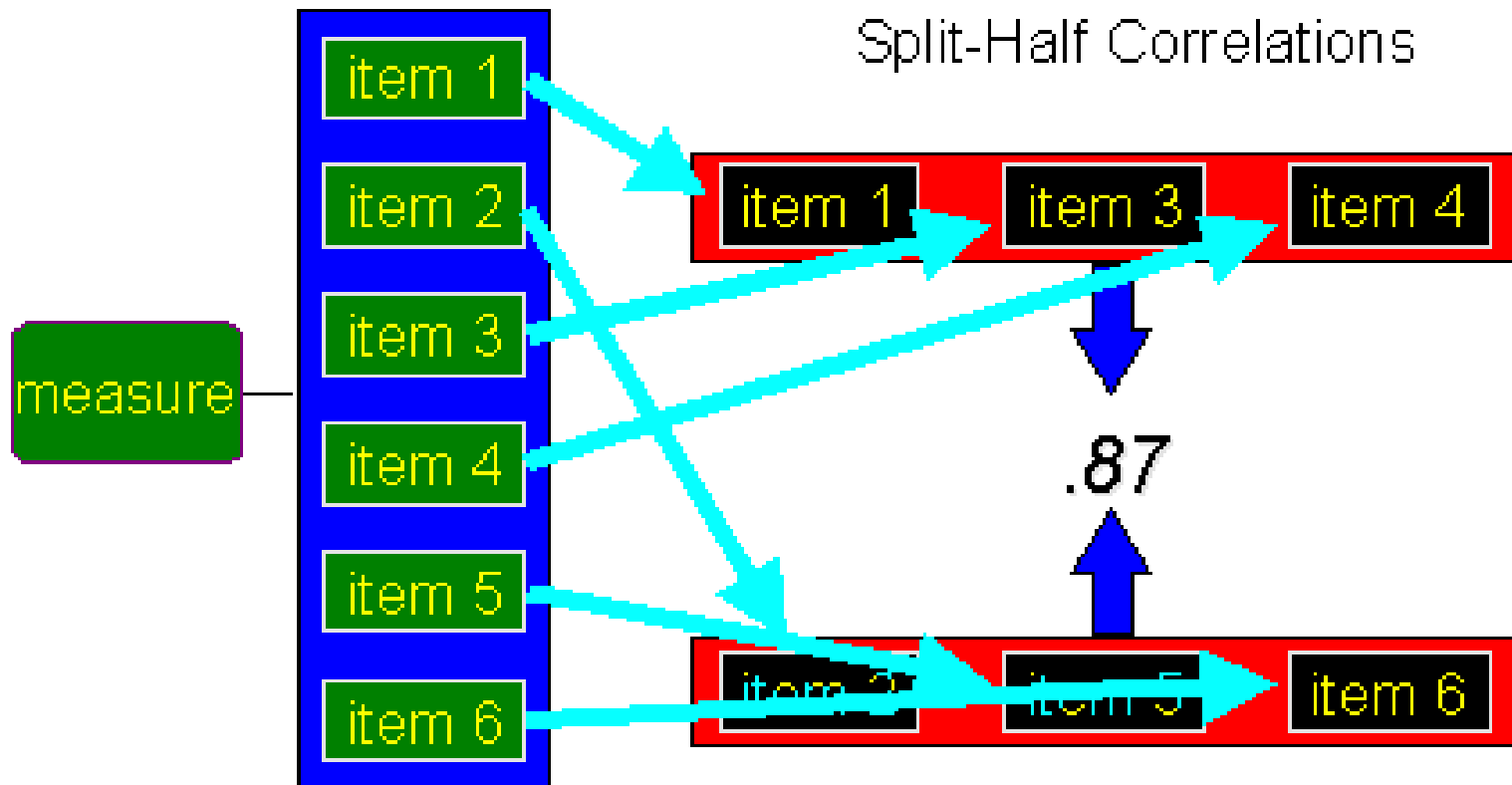
Then I lost some of the questionnaires. No problem: the students cheerfully filled out identical, duplicate questionnaires.

A day later I found the original questionnaires. I compared to two sets to see how the data compared. To my shock, the estimates of study time were completely different! A student who put "4 hours" on the first questionnaire might put "1 1/2 hours" on the second. These two answers came from the *same* student, asked about the *same* chapter (the previous week's assignment) on *two successive days* ! My operational definition of study time, based on the student's retrospective self-report, was clearly unreliable. Therefore it was not valid either; it was not measuring what I thought it measured.

Reliability

- Reliability
 - Consistency of measurements
- Different types of reliability
 - Test reliability
 - Test-retest reliability
 - Internal reliability
 - Split-half reliability

Reliability



Reliability

- Reliability
 - Consistency of measurements
- Different types of reliability
 - Test reliability
 - Test-retest reliability
 - Internal reliability
 - Split-half reliability
 - Cronbach's alpha

Reliability

Tentamen: Introduction to Psychology Engels
Docent: Donk
Faculteit: FGB

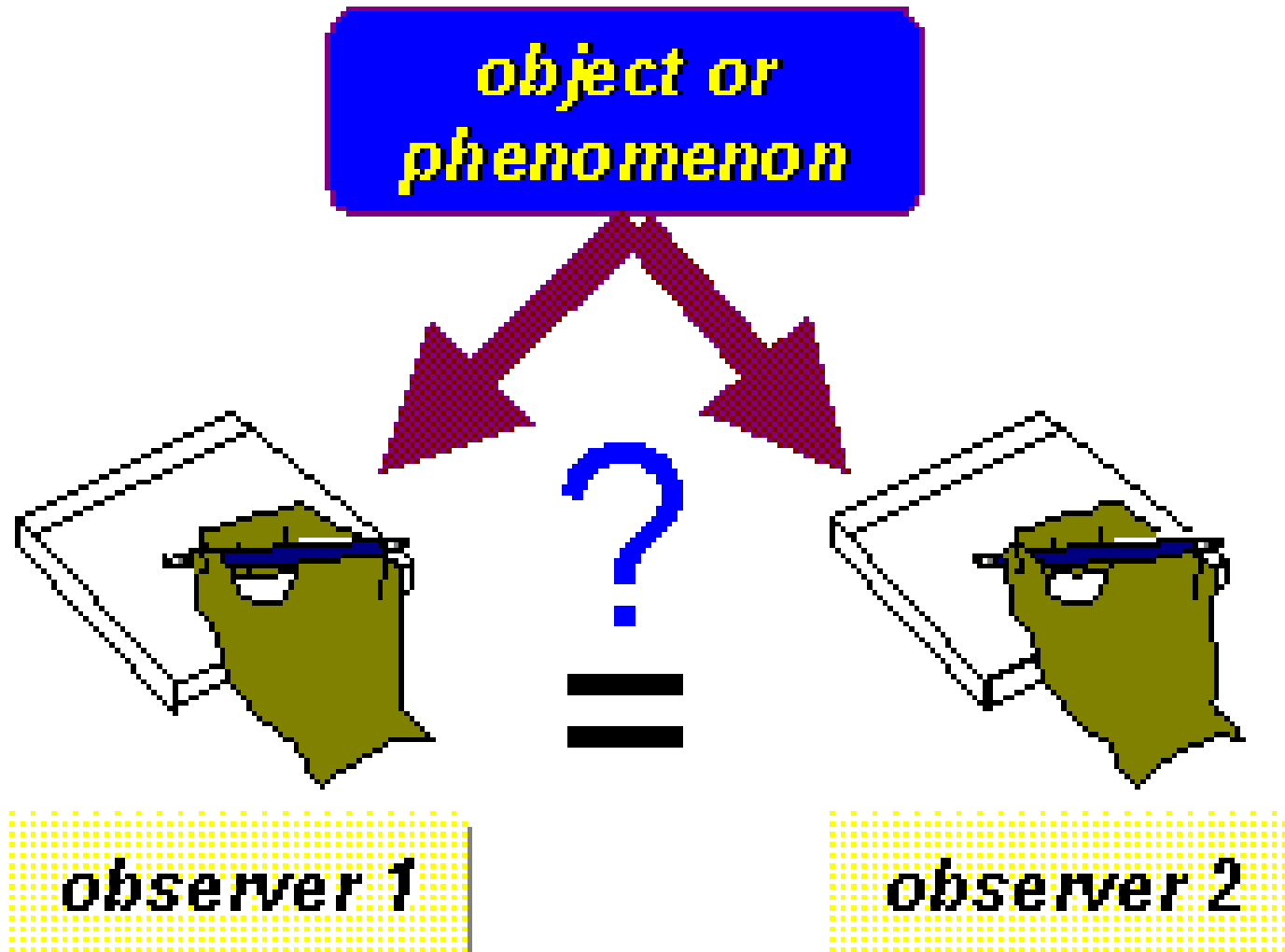
Toetsdatum: 23-10-2017
Afdeling: Cogn. Psych.

(KR20) Fidelity test	0.84
(KR20-75) Fidelity test	0.88
P-avg	0.71
P'-avg	0.62
Average Score	40.00
Standard deviation	7.56
Gemiddelde cijfer:	6.41
Standaardmeetfout:	3.00
Standaard Afwijking cijfer:	1.70
Aantal vragen:	56

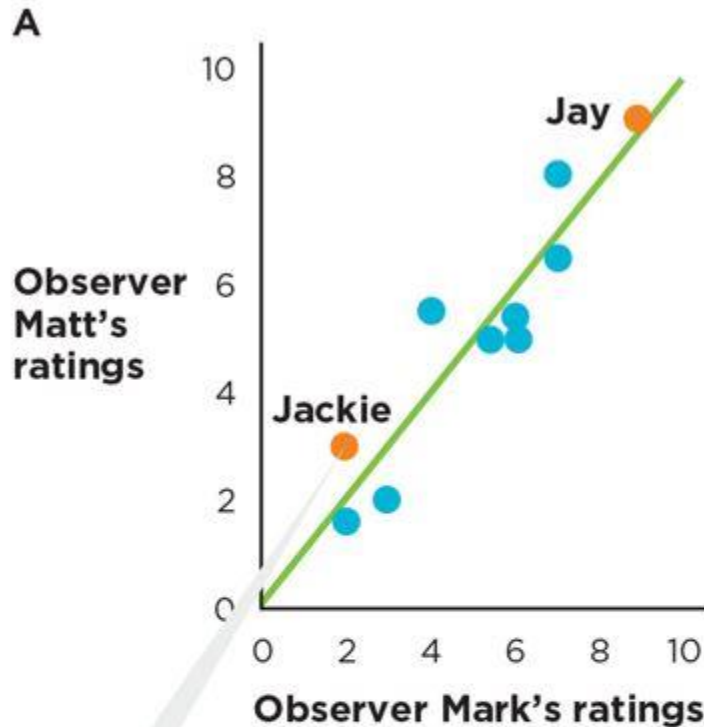
Reliability

- Reliability
 - Consistency of measurements
- Different types of reliability
 - Test reliability
 - Test-retest reliability
 - Internal reliability
 - Split-half reliability
 - Cronbach's alpha
 - Interrater reliability

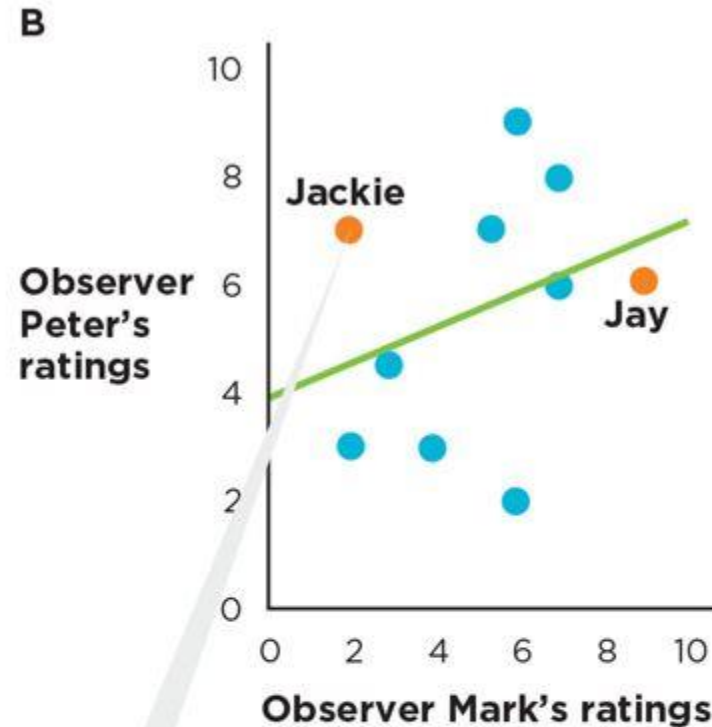
Reliability



Reliability



If the data show this pattern, it means Matt and Mark have good interrater reliability. Mark rated Jackie as one of the least happy children in the sample, and so did Matt. Mark rated Jay as one of the happiest children in the sample, and so did Matt.



If the data show this pattern, it means Mark and Peter have poor interrater reliability. For example, they disagree about Jackie—Mark rated Jackie as one of the least happy children in the sample, but Peter rated her as one of the happiest.

FIGURE 5.3

Reliability

Witness Reports On 9/11

Libby Clark: *"...I believe it was a **passenger plane**, it was relatively large, I mean it **wasn't the size of a jet**, but...er...it was **not a small, small plane**. Like a [unrecognisable]." (From a live ABC / 7 news report at Tuesday 11 September 2001 recorded prior to 9:03 am)*

Reliability

Witness Reports On 9/11

Karim Arraki: *“I saw it come up from the left, and **I saw the plane** coming through to the building, go inside, **a small plane....no, no, it was plane**, you know, like they teach the people to pilot plane, **small plane**, you know, it was that kind of plane....yes, going into the building, and **I never saw that plane before**. It's like something, I don't know, it's like they work with the motors, **I never saw a plane like that before!**”*
(From an ABC News Special Report at 9:08 am on Tuesday 11 September 2001.)

Reliability

Witness Reports On 9/11

Stuart: *“I was waiting a table and I literally saw a, it seemed to be a **small plane**. I just heard a couple of noises, it looked like it like ‘bounced’ of the building and then I heard a, I just saw a huge like ball of fire on top and then the smoke seemed to simmer down....it just seemed like a **smaller plane**, I don’t think it was anything commercial...”* (From a Bryant Gumble CBS News Special Report at 8:52 am on Tuesday 11th September 2001).

Reliability

Witness Reports On 9/11



Reliability

- Reliability
 - Consistency of measurements
- Different types of reliability
 - Test reliability
 - Test-retest reliability
 - Internal reliability
 - Split-half reliability
 - Cronbach's alpha
 - Interrater reliability
 - Experimental reliability

Descriptive research

- Observational research
 - The process of watching people or animals and systematically recording how they behave or what they are doing
- Survey research
 - A method of posing questions to people on the telephone, in personal interviews, on written questionnaires, or via the internet
- Case studies

Goals and Tools of Psychology:

Descriptive Methods

To **describe** human and animal behavior and mental processes

- Observations (naturalistic or lab)
- Surveys
- Case study



MICHAEL NICHOLS/National Geographic Creative



FIGURE 1.2 Twitter message moods, by time and by day

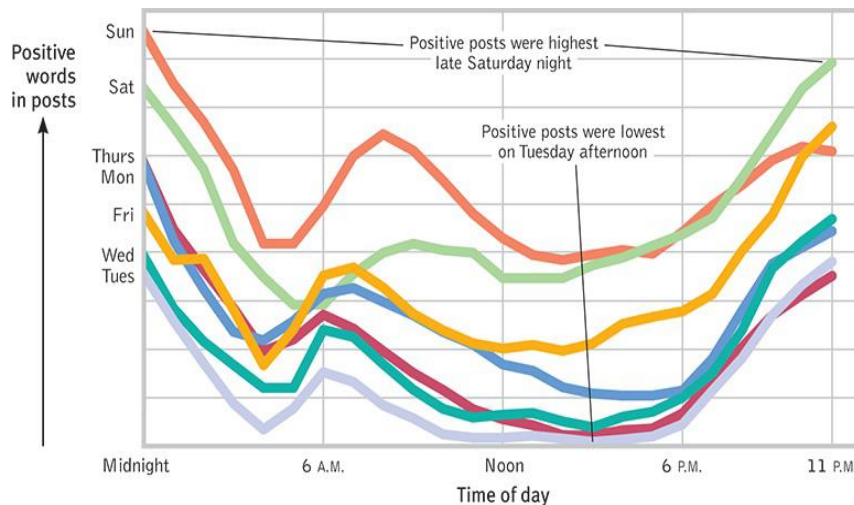


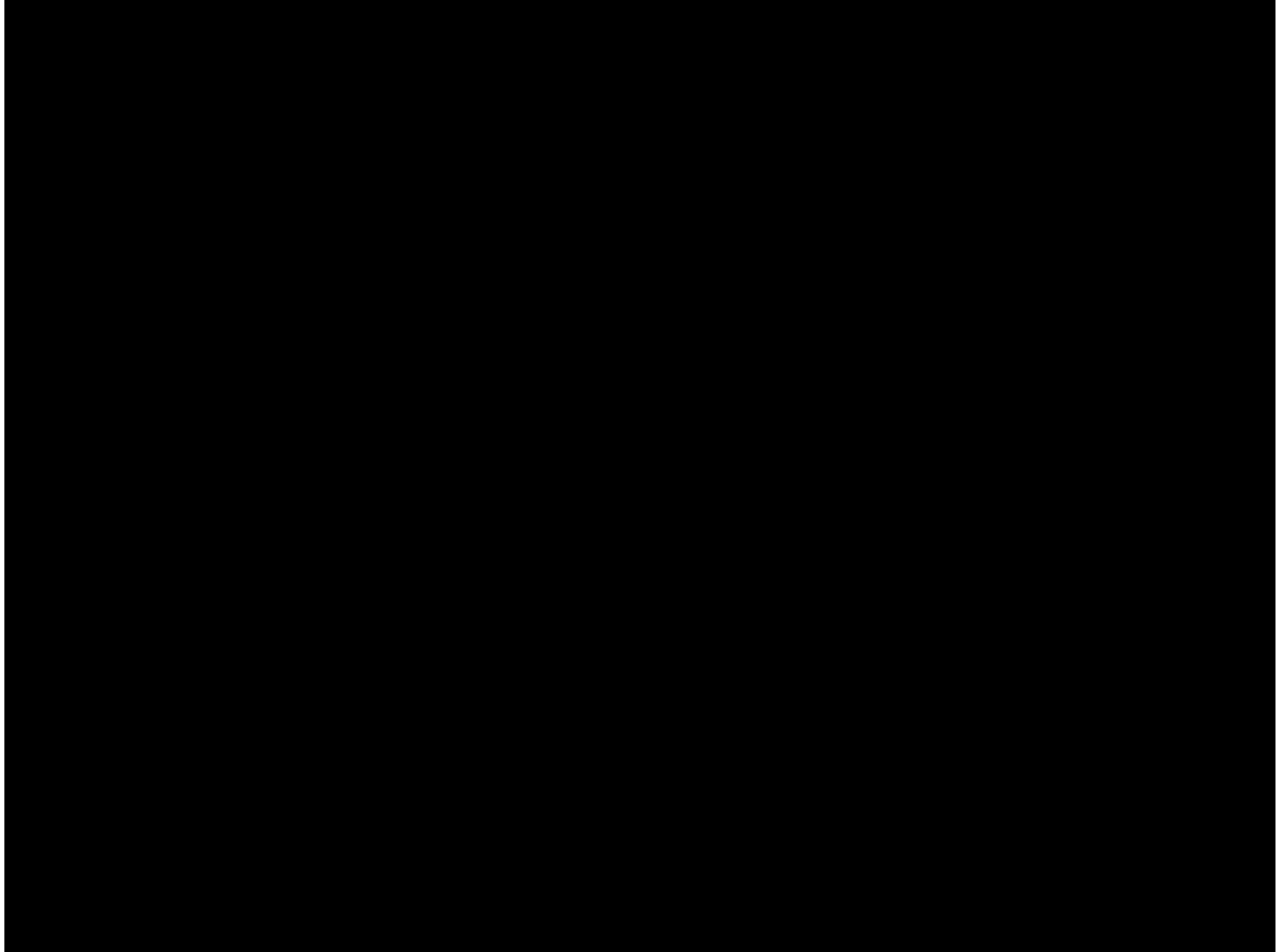
FIGURE 1.2 Myers/DeWall, *Psychology*, 12e, © 2018 Worth Publishers
Data from Golder & Macy, 2011.



Descriptive research

- Observational research
 - The process of watching people or animals and systematically recording how they behave or what they are doing
 - to describe nature
 - to generate hypotheses (not to test hypotheses!)
 - e.g., *Bystander effect*

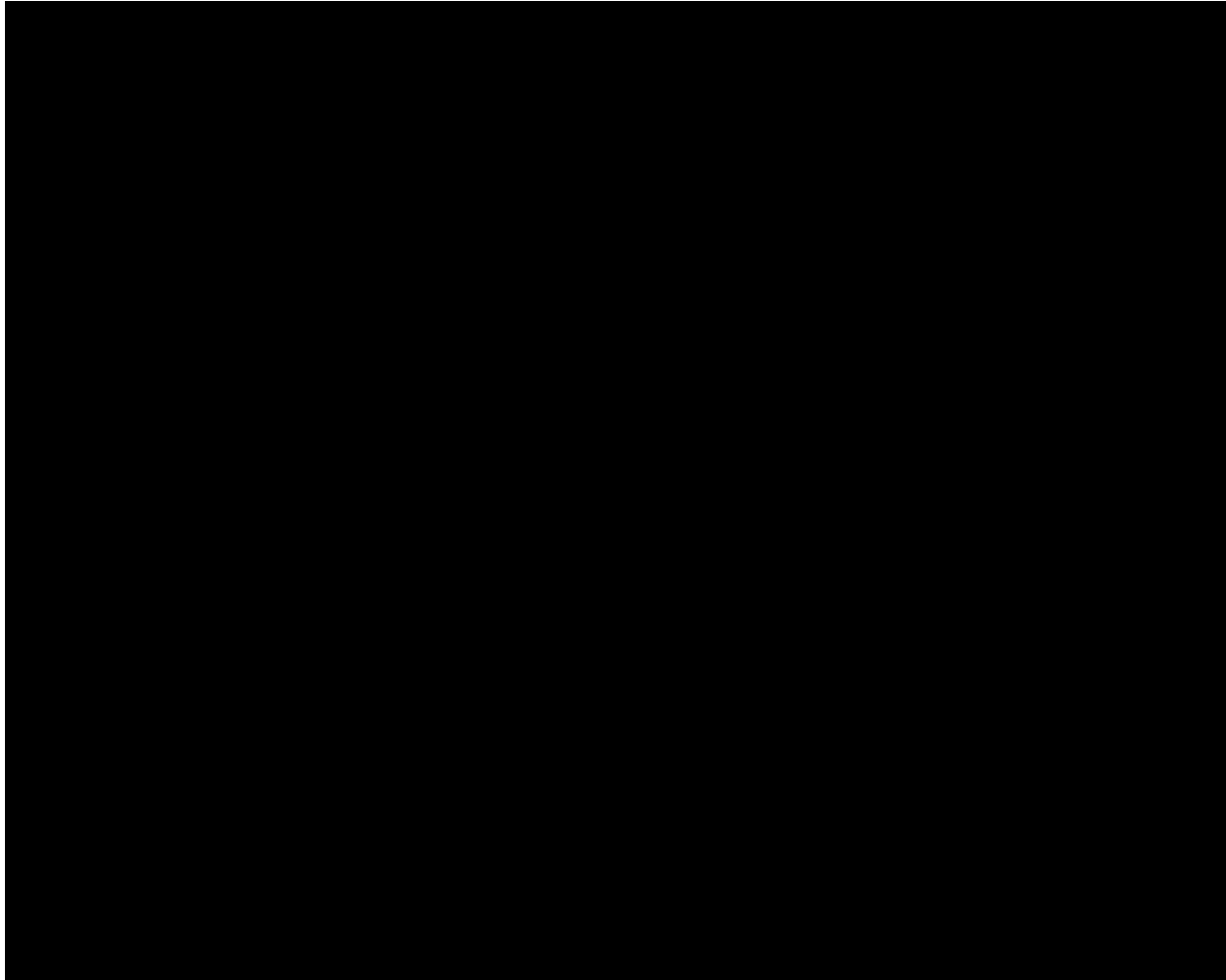
Descriptive research



Descriptive research



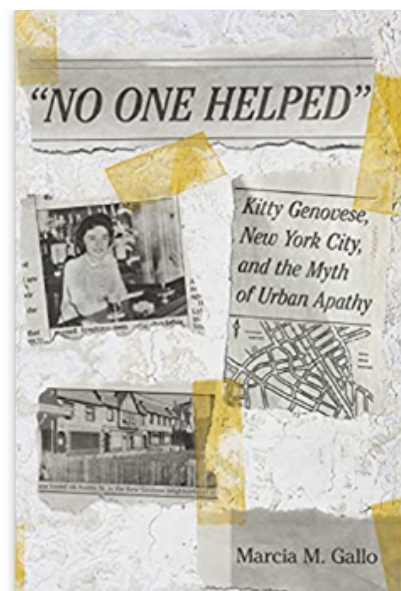
Descriptive research



"No One Helped": Kitty Genovese, New York City, and the Myth of Urban Apathy Paperback – August 11, 2015

by [Marcia M. Gallo](#) (Author)

★★★★☆ 20 ratings



"No One Helped" places the conscious creation and promotion of the Genovese story within a changing urban environment. Gallo reviews New York's shifting racial and economic demographics and explores post-World War II examinations of conscience regarding the horrors of Nazism. These were important factors in the uncritical acceptance of the story by most media, political leaders, and the public despite repeated protests from Genovese's Kew Gardens neighbors at their inaccurate portrayal. The crime led to advances in criminal justice and psychology, such as the development of the 911 emergency system and numerous studies of bystander behaviors. Gallo emphasizes that the response to the crime also led to increased community organizing as well as feminist campaigns against sexual violence. Even though the particulars of the sad story of her death were distorted, Kitty Genovese left an enduring legacy of

The responsive bystander and group intervention

Mark Levine

Affiliations

PMID: 19025
DOI: 10.1037

 Full text

Abstract

Four experiments examined bystander behavior in a group intervention. In addition, groups were encouraged to help examine the greater help from women

> Proc Natl Acad Sci U S A. 2017;114(18):9583-9588. doi: 10.1073/pnas.1817100114

Toddlers and infants helping an ingroup member

Fransisca Ting¹, Zijjing

Affiliations

PMID: 30858320
PMCID: PMC6442622
DOI: 10.1073/pnas.1817100114

Free PMC article

 Full text links

Abstract

Adults and older children help a transgressor when the violation-of-expectation toddlers (Exps. 1 and 2) an individual in a minority to an ingroup victim. We To start, children saw a Next, the wrongdoer ne hindered her. The group relative to that of the b 2) or novel outfits (Exps. 3 and 4). When the victim belonged to a bystander, children expected TPP: At both ages, they detected the bystander chose to help the wrongdoer. Across experiments,

> Exp Psychol. 2021 Mar;68(2):107-117. doi: 10.1037/xap0000328

Be Responsible?

Brian P Meier¹, Michael B Kitchens¹, Nevada S Keyton¹, Sydney E Petra¹, Matthew M Simmers¹, John O Unc

Affiliations

PMID: 34405691
DOI: 10.1027/1618-3169/a000513

 Full text links

Abstract

The bystander effect reveals that people are less likely to help others when others are present. We examined the bystander effect in a field study assigned to a two (Bystanders: none or two) and two (Shirt: blank shirt and shirt with "Be Responsible") design. The researcher dropped eight pens appearing to be broken while wearing one of the two shirts in the field. The bystander effect was found: People were less likely to help in the no bystanders condition (59.05%) than in the two bystanders condition (41.67%) when the researcher was wearing the "Be Responsible" shirt. Helping rates, but it did not significantly differ as part of an interaction term. Thus, priming the concept of responsibility

> J Interpers Violence. 2021 Dec;36(23-24):11532-11552. doi: 10.1177/0886260519900328. Epub 2020 Jan 15.

Empathy Is Not Enough: The Inhibiting Effects of Rape Myth Acceptance on the Relation Between Empathy and Bystander Intervention

Ruschelle M Leone¹, Kristen N Oyler¹, Dominic J Parrott¹

Affiliations

PMID: 31941405
DOI: 10.1177/0886260519900328

 Full text links

Abstract

Many bystander training programs aim to build empathy and decrease false and often prejudicial beliefs about rape and its occurrence (i.e., rape myth acceptance) to encourage prosocial bystander intervention. However, little empirical research has examined the association between these constructs. The aim of the present study was to examine the effect of rape myth acceptance on the relation between both (a) the ability to experience and share the emotions of others (i.e., emotional empathy) and (b) the ability to understand the emotions of others (i.e., cognitive empathy), and bystander decision-making and intentions to intervene. In Study 1, 154 undergraduate men completed measures of empathy, rape myth acceptance, and bystander decision-making. Findings demonstrated that emotional empathy was associated with more perceived pros for intervention among men with low, but not high, levels of rape myth acceptance. In Study 2, 185 undergraduate men and 210 undergraduate women completed similar measures as in Study 1, with the addition of a measure on bystander intentions to intervene. Results detected a conditional indirect effect of emotional and cognitive empathy on bystander intentions to intervene via the perception that there are more pros than cons to intervention. Findings suggest that empathic individuals may weigh helping a potential victim (i.e., pro) over hurting their own self-image (i.e., con). However, this process is inhibited among men who endorse greater rape myths, likely because they are inhibited from intervening at various steps of the decision-making model. Collectively, these results highlight the importance of

Descriptive research

- Observational research
 - Problems
 - People are not always good at observation
 - This is a threat to the interrater reliability
 - Low internal validity

Descriptive research

- Observational research
 - Problems
 - People are not always good at observation
 - This is a threat to the interrater reliability
 - Low internal validity
 - Threats to construct validity
 - Observer bias

Descriptive research



Descriptive research

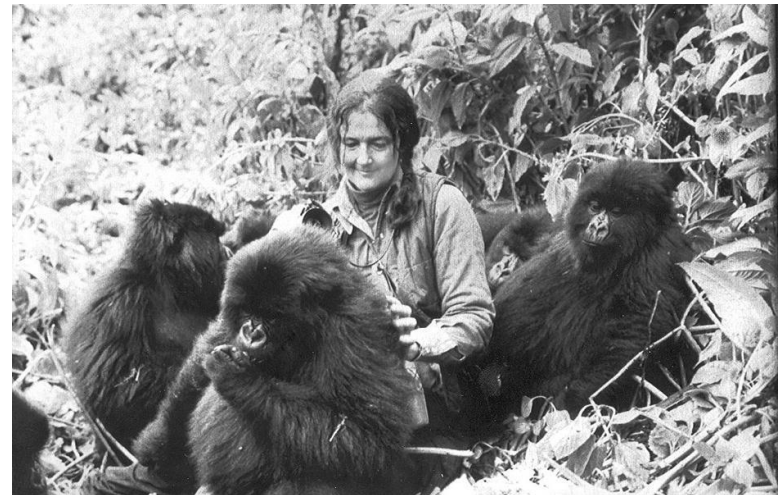
- Observational research
 - Problems
 - People are not always good at observation
 - This is a threat to the interrater reliability
 - Low internal validity
 - Threats to construct validity
 - Observer bias
 - Observer effects (towards observers expectation)
 - Reactivity (no particular direction)
 - unobtrusive observations

Descriptive research



Descriptive research

- Observational research
 - Problems
 - People are not always good at observation
 - This is a threat to the interrater reliability
 - Low internal validity
 - Threats to construct validity
 - Observer bias
 - Observer effects
 - Reactivity
 - unobtrusive observations
 - "wait it out" strategy (participant observation)



Descriptive research

- Observational research
 - Problems
 - People are not always good at observation
 - This is a threat to the interrater reliability
 - Low internal validity
 - Threats to construct validity
 - Observer bias
 - Observer effects
 - Reactivity
 - unobtrusive observations
 - "wait it out" strategy (participant observation)
 - unobtrusive measures



Descriptive research

- Survey (or poll)
 - A method of posing questions to people on the telephone, in personal interviews, on written questionnaires, or via the internet
 - Question formats
 - Open-ended questions
 - *What do you think of this class?*
 - Forced-choice format
 - *Do you like this class so far? Yes or No*
 - Likert scale
 - *I like this class*

1	2	3	4	5
strongly disagree			strongly agree	
 - Semantic differential format
 - *How difficult is this class?*

1	2	3	4	5
very easy			very difficult	

Descriptive research

- Survey (or poll)
 - Problems
 - Low internal validity
 - Threats to construct validity
 - Question wording
 - Leading question
 - *How fast do you think the car was going when he hit the other car?*
 - *How fast do you think the car was going when he smashed into the other car?*
 - Double-barreled question
 - *Do you enjoy swimming and wearing sunscreen?*
 - Negatively worded question
 - *People who do not drive with a suspended license should never be punished.*
Disagree 1 2 3 4 5 Agree
 - Question order

Descriptive research

- Survey (or poll)
 - Problems
 - Low internal validity
 - Threats to construct validity
 - The use of response sets (e.g., *yea-saying*, *fence sitting*)
 - Trying to look good or bad (faking good or bad)
 - People are not always capable of reporting accurately about their feelings, thoughts, and behavior!

Descriptive research

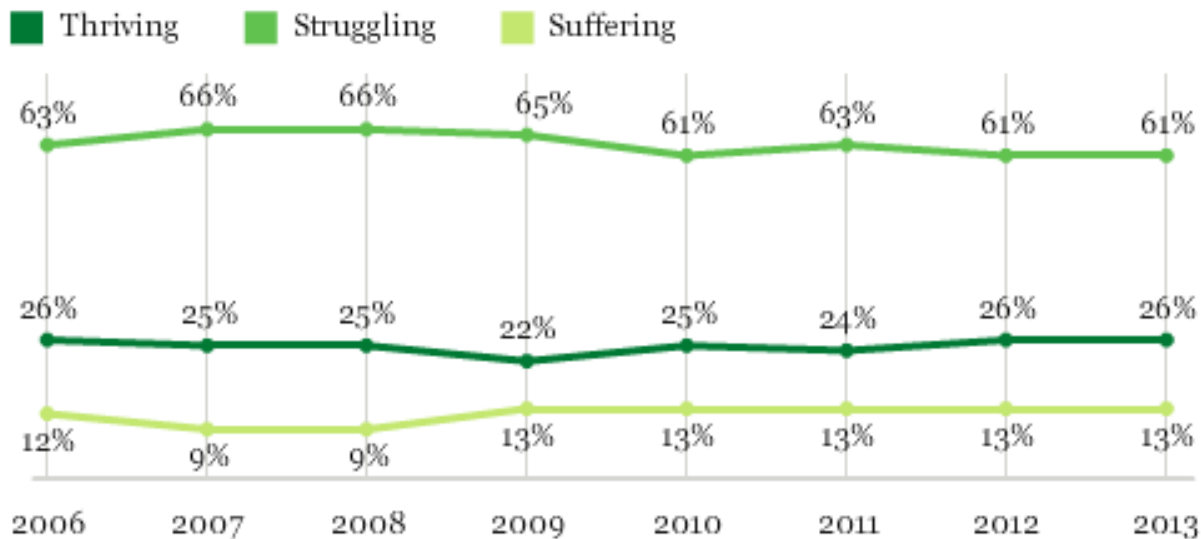
- Descriptive research → frequency claims
 - Threats to external validity
 - Biased or unrepresentative sample

Descriptive research

- Descriptive research → frequency claims
 - "Three in four women worldwide rate their lives as "struggling" or "suffering" (Gallup, 2015)

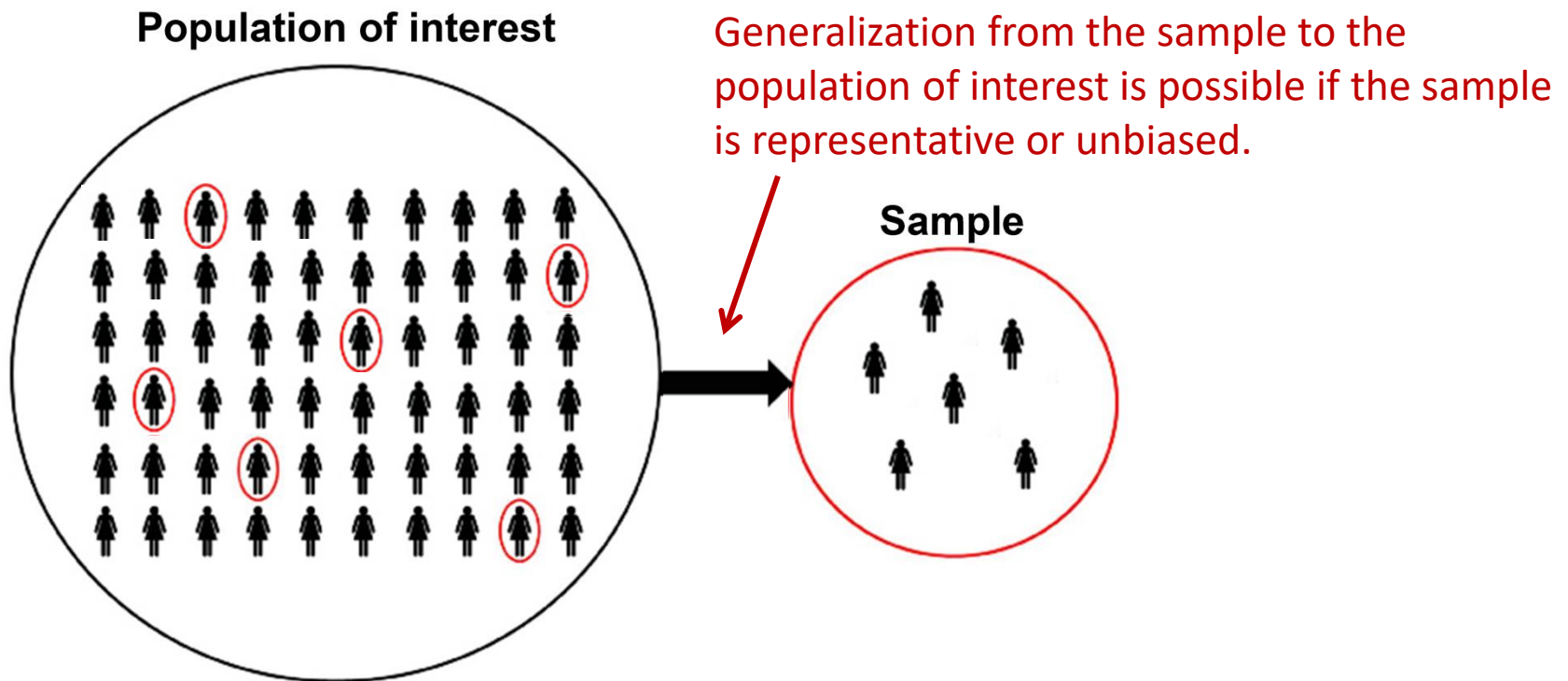
Life Evaluation -- Women

The Life Evaluation Index measures respondents' perceptions of where they stand now and in the future.



Descriptive research

- Descriptive research → frequency claims
 - "Three in four women worldwide rate their lives as "struggling" or "suffering" (Gallup, 2015)



Descriptive research

- Descriptive research → frequency claims
 - Threats to external validity
 - Biased or unrepresentative sample
 - Convenience:
 - studies may use samples consisting of only those who are easy to contact
 - Self-selection:
 - many studies use samples consisting of only those who volunteer
 - To obtain a representative sample there are various techniques
 - random sample
 - different rewards
 - replication
 - Threats to statistical validity
 - A too small sample size

Descriptive research

TABLE 7.4

Margins of Error Associated with Different Random Sample Sizes

IF THE PERCENTAGE IS ESTIMATED ON A RANDOM SAMPLE OF SIZE	MARGIN OF ERROR ON THE PERCENTAGE IS
2,000	Plus or minus 2%
1,500	Plus or minus 3%
1,000	Plus or minus 3%
500	Plus or minus 4%
200	Plus or minus 7%
100	Plus or minus 10%
50	Plus or minus 10%

Note: Margin of error varies as a function of sample size and the percentage result of the poll. In this table, estimates were based on a 50% polling result (e.g., if 50% of people supported a candidate).

Descriptive research

- Descriptive research → frequency claims
 - The internal validity of frequency claims is low
 - It is important to question how well
 - each variable was measured (construct validity)
 - the results generalize (external validity)
 - the numbers support the conclusions (statistical validity)
 - reliability

Descriptive research

HEADLINES

SHARE THIS



US polls show Clinton ahead but battleground states up for grabs

Agence France-Presse / 02:31 AM November 07, 2016

WASHINGTON — Just two days before the bitter US presidential election, national polls gave the edge to Hillary Clinton, with results too close to call in some key battleground states where the race will be won or lost.

